

Ives Sanjines Iriarte

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Orlando, FL - Bilingual in English and Spanish

Education

University of Central Florida (UCF)

Bachelor of Science in Mechanical Engineering | GPA: 3.5

Orlando, FL

Expected: Dec 2027

Projects

Scuderia SHPE - AI Racer Competition | Mechanical & Integration Project Lead

Spring 2026

- **Led a team of 15 engineering students** as the **Mechanical & Integration Lead**, owning the full mechanical design lifecycle from concept to competition-ready vehicle.
- **Reverse engineered** a chassis from an STL scan in **SolidWorks**, producing detailed 3D models and assemblies used to design custom mounting brackets and enclosures for sensor and compute hardware.
- Packaged and integrated electronic modules (Raspberry Pi, LiDAR, wiring harness) into mechanical enclosures, managing routing constraints and component fit across the assembly.
- Fabricated a laser-cut top chassis and 3D-printed bottom structure, supporting assembly through soldering and welding.
- Collaborated cross-functionally with project coordinators and sub-teams, incorporating weekly feedback to drive design improvements within budget constraints.

PowerTrain AI | SolidWorks, Schematics, ESP32, C/C++, Soldering, Python, PyTorch

Fall 2025

- **1st place winner** of the SHPE UCF Projects Competition Fall 2025 for developing an autonomous driving car.
- Designed chassis in **SolidWorks**, producing detailed part drawings and assemblies, and maintained a **Bill of Materials (BOM)** tracking all mechanical and electronic components across design using Microsoft Excel.
- Integrated PCBs, wiring, and electronic modules into the mechanical enclosure, performing fit checks and tolerance studies to ensure reliable assembly.
- Implemented a **PyTorch neural network** to classify **10** driving levels for autonomous driving battery optimization.
- Collected **1,000+ telemetry samples** to train the ML model, capturing test results and observations that improved battery efficiency by **2–3× longer runtimes**.

Professional Experience

Mechanical Properties & Characterization Lab @ UCF

March 2026 – Present

Undergraduate Research Assistant

Orlando, FL

- Member of the **Rig Team**, designing, fabricating, and assembling custom test rigs to support faculty-led research in mechanical properties and material characterization.
- Conducted tensile testing on **MTS mechanical testing machines** following **ASTM standards**, preparing specimens and recording data for material behavior analysis.
- Currently rebuilding a **column buckling machine** originally used by the U.S. Air Force to test materials under hypersonic conditions integrating **microcontrollers** to control **load cells**, **LVDT sensors**, and a **stepper motor** for modernized, automated testing.

Bolivian Student Association @ University of Central Florida

Aug 2025 – Feb 2026

Co-Founder Treasurer

Orlando, FL

- Co-founded and served as **Treasurer** for a new cultural RSO supporting Bolivian students at UCF; established the organization's financial framework and operational processes with other founder members.
- Built and maintained a **full financial tracking system using Excel** (budgets, ledgers, reimbursements), overseeing **100%** of organizational transactions with accurate documentation and audit-ready records.
- Coordinated logistics and budgeting for **3–5** cultural and community events per semester, increasing organization outreach.

Aramark @ University of Central Florida

August 2025 – Present

Student Worker

Orlando, FL

- Supported daily food preparation and service, following strict safety, sanitation, and HACCP standards.
- Operated commercial kitchen equipment and maintained workstation cleanliness during high-volume service periods serving **500–1,000+** students per shift.

Skills

Design: SolidWorks, Assemblies, Ansys.

Manufacturing: 3D Printing, Soldering, Laser Cutting, Schematics, Circuits, Assembly, Sensors.

Documentation: Bill of Materials (BOM), Technical Reports, .

Programming/Computer: MATLAB, C, Arduino/ESP32, Python, LaTeX, Windows, Office365, Excel.

Soft Skills: Teamwork, Problem-Solving, Adaptability, Leadership, Critical Thinking, Conflict Resolution.